LESSON 3.7

Ratio Reasoning: Convert Measurements Between Systems

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use ratio reasoning to convert measurements between the customary and metric systems.

WHY THIS MATTERS The academy's kitchen orders from suppliers in two different countries. One ships in pounds and quarts, the other in kilograms and liters. Compare measurements by converting with the appropriate factor. Some factors are exact; rounded factors give approximate results. One supplier ships 10 pounds of rice and the other ships 5 kilograms. Use 1 pound ≈ 0.45 kilogram to find which shipment is more.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Conversion factor *Spanish: Factor de conversión*	The ratio that links a unit in one system to a unit in the other, such as 1 mile to about 1.609 kilometers.	1 m ≈ 3.28 ft
Customary system *Spanish: Sistema usual*	The measurement system used in the United States: inches, feet, miles, ounces, pounds, cups, quarts, gallons.	in · ft · mi · oz · lb · qt · gal
Metric system *Spanish: Sistema métrico*	The measurement system used in most of the world, built on tens: centimeters, meters, kilometers, grams, kilograms, liters.	cm · m · km · g · kg · mL · L
Approximately *Spanish: Aproximadamente*	Close to, but not exactly equal. A rounded conversion factor gives an approximate result, written with ≈; an exact conversion uses =.	≈ means “about”
Ratio table *Spanish: Tabla de razones*	A table of equal ratios used to scale a conversion factor up to the amount you actually need.	km: 1, 10, 100 · mi: 0.6, 6, 60
Reasonableness *Spanish: Sentido lógico*	Checking that the size of an answer makes sense for the situation.	bigger unit → smaller number

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A bag weighs 44 pounds. About how many kilograms? (1 kg ≈ 2.2 lb)

1. A kilogram is heavier, so the number gets smaller: divide.
2. 44 ÷ 2.2 = 20.
3. Check: 20 × 2.2 = 44.

Answer: 44 lb ≈ 20 kg

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Terrance needs flea medicine for a dog weighing 31 pounds, but the packages are labeled in kilograms.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I know 1 _____ ≈ _____ _____ .
- A _____ is bigger than a _____ , so I _____ .
- My answer is about _____ _____ .
- This is reasonable because _____ .

Word bank convert | conversion factor | approximately | about | reasonable | bigger | smaller | multiply | divide | meter | mile | kilogram

WATCH OUT!

A common mistake

Compare measurements only after converting to a common unit. Also check whether the conversion factor is exact or rounded: 1 inch = 2.54 cm and 1 international mile = 1.609344 km are exact

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Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

1 kilometer ≈ 0.6 mile. Complete the ratio table.

Kilometers	Miles
1	0.6
10	
100	

HINT Row 1 is the conversion factor.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

1 kilometer ≈ 0.6 mile. About how many miles is 5 kilometers?

(A) 3 miles
(B) 5.6 miles
(C) 8 miles
(D) 30 miles

HINT Write the conversion factor as a ratio: 1 : 0.6.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

1 pound ≈ 0.45 kilogram. About how many kilograms is 10 pounds?

(A) 4.5 kg
(B) 10.45 kg
(C) 22 kg
(D) 45 kg

HINT The ratio is 1 lb : 0.45 kg.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which pair of units comes from two DIFFERENT measurement systems?

(A) Inches and feet
(B) Cups and quarts
(C) Grams and kilograms
(D) Miles and kilometers

HINT Sort each unit as customary or metric.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A sign in Mexico gives a distance of 80 kilometers. Using 1 km ≈ 0.6 mi, about how far is that in miles?

(A) About 8 miles
(B) About 48 miles
(C) About 80 miles
(D) About 133 miles

HINT Scale the ratio 1 : 0.6 up to 80 kilometers.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Setup** 1 kilometer ≈ 0.6 mile. About how many miles is 20 kilometers?
2. **Step 1** The ratio is 1 : 0.6.
3. **Step 2** 20 ÷ 0.6 = 33.3
4. **Step 3** 20 kilometers ≈ 33.3 miles.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Is a kilometer longer or shorter than a mile?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why does 1 inch = 2.54 cm use =, while 1 mile ≈ 1.609 km uses ≈? What does the rounding change?

SENTENCE STARTER The inch factor is ___ by definition. The mile factor shown is ___, so I write ___ for its result.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.7

Ratio Reasoning: Convert Measurements Between Systems

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

1 pound ≈ 0.45 kilogram. Complete the ratio table, then use it to place a 31-pound dog.

Pounds	Kilograms
1	0.45
10	
30	
40	

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Converting between systems: sort each conversion by what happens to the number.

Groups: The number gets smaller | The number gets bigger

_____ Kilometers → miles

_____ Miles → kilometers

_____ Pounds → kilograms

_____ Kilograms → pounds

_____ Inches → centimeters

_____ Centimeters → inches

3 **Mark and label each value on the number line.**

Using 1 km ≈ 0.6 mi, place the mile markers that match 0, 50, and 100 kilometers.



SHOW YOUR THINKING

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A dog weighs 31 pounds. Flea medicine comes in packages for 4–10 kg, 10–25 kg, and 25–40 kg. Which package is correct?

(A) The 4–10 kg package
(B) The 10–25 kg package
(C) The 25–40 kg package
(D) None — 31 is above every range

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

One of the academy's suppliers ships olive oil in 4-liter tins. Using 1 liter ≈ 1.06 quarts, about how many quarts of oil are in one tin?

(A) About 3.77 quarts
(B) About 4.24 quarts
(C) About 5.06 quarts
(D) About 42.4 quarts

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Setup** A recipe calls for 2 pounds of flour. The scale is metric. (1 lb ≈ 0.45 kg)
2. **Step 1** The conversion factor is 1 : 0.45.
3. **Step 2** 0.45 × 2 = 0.9, so 2 pounds ≈ 0.9 kilogram.
4. **Step 3** 2 pounds = 0.9 kilogram exactly.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why does 1 inch = 2.54 cm use =, while 1 mile ≈ 1.609 km uses ≈? What does the rounding change?

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LESSON 3.7

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Challenge · Extension

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- ☐ Meets target
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the conversion table. The inch factor is exact. Round the other metric factors to three decimal places.

Customary	Metric equivalent
1 inch	2.54 cm (exact)
1 mile	
1 pound	
1 United States liquid quart	

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 **Find the mistake. Explain it, then show the correct work.**

Find the error

1. **Setup** Which is longer, 5 kilometers or 5 miles? (1 km ≈ 0.6 mi)
2. **Step 1** Both numbers are 5.
3. **Step 2** The units are different, so I compare the numbers: they are equal.
4. **Step 3** 5 kilometers is the same distance as 5 miles.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 **Answer in complete sentences.**

The Golden Gate Bridge can hold about 35,924,000 pounds. An average midsize car weighs about 1,524 kilograms. Using 1 kilogram ≈ 2.2 pounds, about how many midsize cars could be on the bridge at one time? Explain each step, and say why your answer is an estimate.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Why does 1 inch = 2.54 cm use =, while 1 mile ≈ 1.609 km uses ≈? What does the rounding change?

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